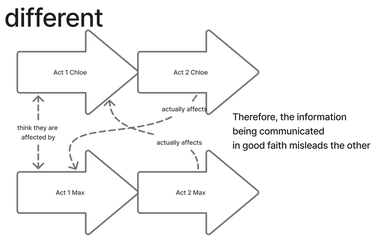
## Character Timeline

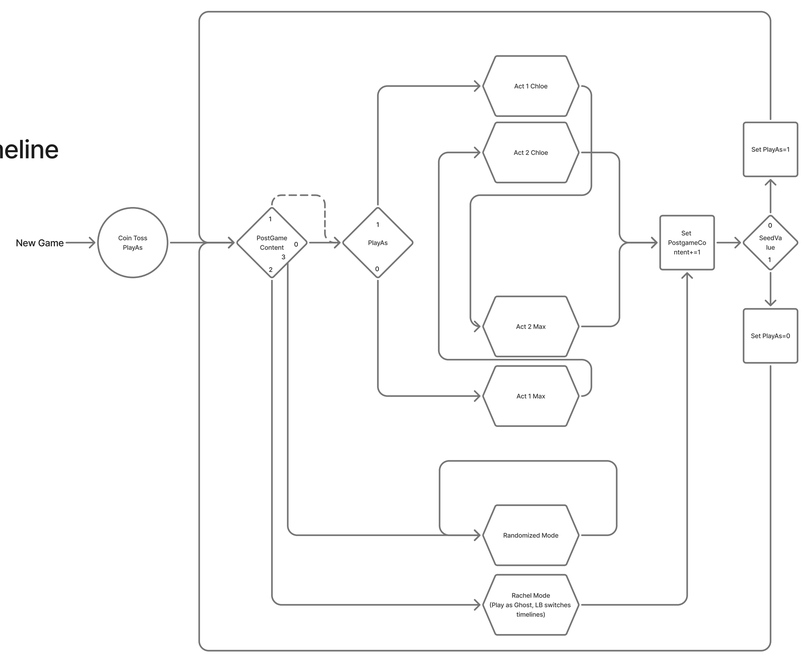
this structure doesn't quite work if timelines are different




Therefore, the information being communicated in good faith misleads the other

There is no "breaking" the cycle, only to learn from your mistakes and do a little better each time

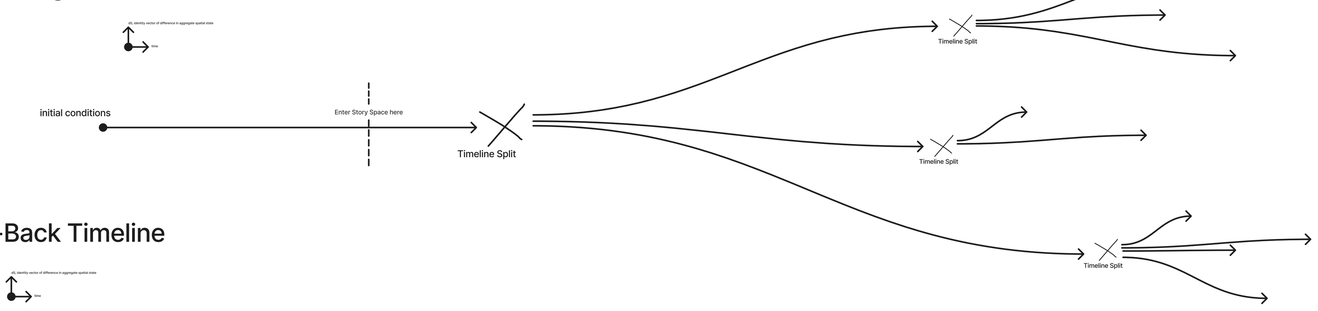
## Player Timeline




# Complex-System Temporal Dynamics

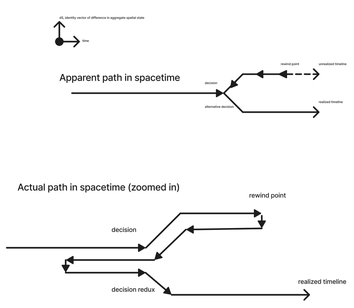
## Branching Timeline

"Branching Timeline" model of the original game assumes that decision points in the timeline divert the path of causality into one of several separate 'lanes' of temporal momentum, where this small perturbation of conditions at the decision point will propagate along and exponentially compound the number of possible end states




## Double-Back Timeline

There are potential alternative interpretations of this behavior, however, that I have chosen to apply. The first I will call the "Double-Back" interpretation, where the rewind is an unbroken chain of causality, sufficiently driven in parallel spatial conditions (though reversed) to the same point in time *t*, but along an almost negligibly different path. The causality *c* impacts the spatial and temporal state over successive *c = c + 1* steps, where causality is **not** synonymous with time *t*. Therefore the causal driver impacts the backwards movement along the time axis, without "overwriting" the original path of causality that has been traversed. The state of the system up to the point of rewind appears close enough to the same, but is not in fact the same state which has already been trodden.

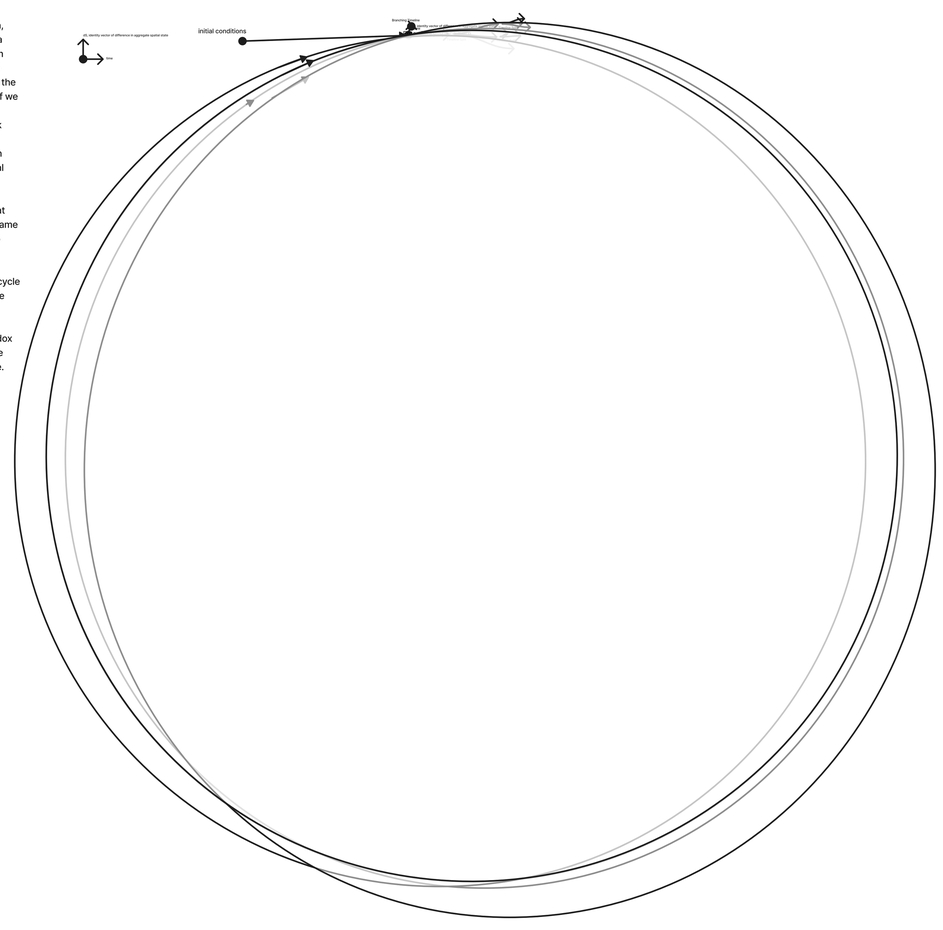



## Orbit Timeline

A different approach, which I will call Orbit interpretation, views the apparent "branching timeline" as a feature of a single causal loop over sufficiently high steps of iteration from the initial conditions. (Zoom out.) This is applicable even if the previous assumption of time decoupled from the chain of causality does not prove to be a valid premise. If we instead frame our system through the lens of a cyclical departure and return to the same initial conditions (think expansion and contraction of the universe), then any "rewind" could be instead achieved by a sufficiently high iteration of step *c* along *t*, in the quantity of one universal cycle minus the desired rewind amount.

The nature of both interpretations comes with the caveat that the traveler never returns to a point of *exactly* the same spatial conditions (even when the differences appear so negligible as to be nonexistent), only *approximately* the same point. Such is the nature of complex dynamical systems. Notably this means that the apparent circular cycle of time is more accurately modeled as a spiral, one of the core visual motifs of Life is Strange.

These philosophy of time travel negate any logical paradox or inconsistency, because any such perturbations by the time traveler do not actually violate the causal sequence.




note: possible self-similar fractal timeline along partial time dimensions, this interpretation also has potential. Fractal could even be a partial double-back within an orbit

# What does this all mean?

some pseudo-philosophical nonsense

You contain multitudes, but you are a reflection of yourself. The whole of you is emergent of the pieces of you. fractals

There are no means that are justified by ends. Not because nothing ever ends, but because the means are the manifestation of the ends.

cycles of behavior are not patterns we break free from after a single decision, a single effort. They are continuous processes which we must inflect attentively and minutely and *consciously* along each step of the path, until the system state gravitates to a different orbit. Conversely, just because you seem to find yourself in the same place does not mean you have not grown. It is not the same place, and you are not the same person. You can't go home again, but also you don't have to. The system state at time *t* will progress to a different state at ~~*t+1*~~ t + dt_cycle than it progressed to on a previous orbit.

Don't push against the current, flow with your environment and the small changes will add up over time until the same apparent conditions produce a dramatically different outcome